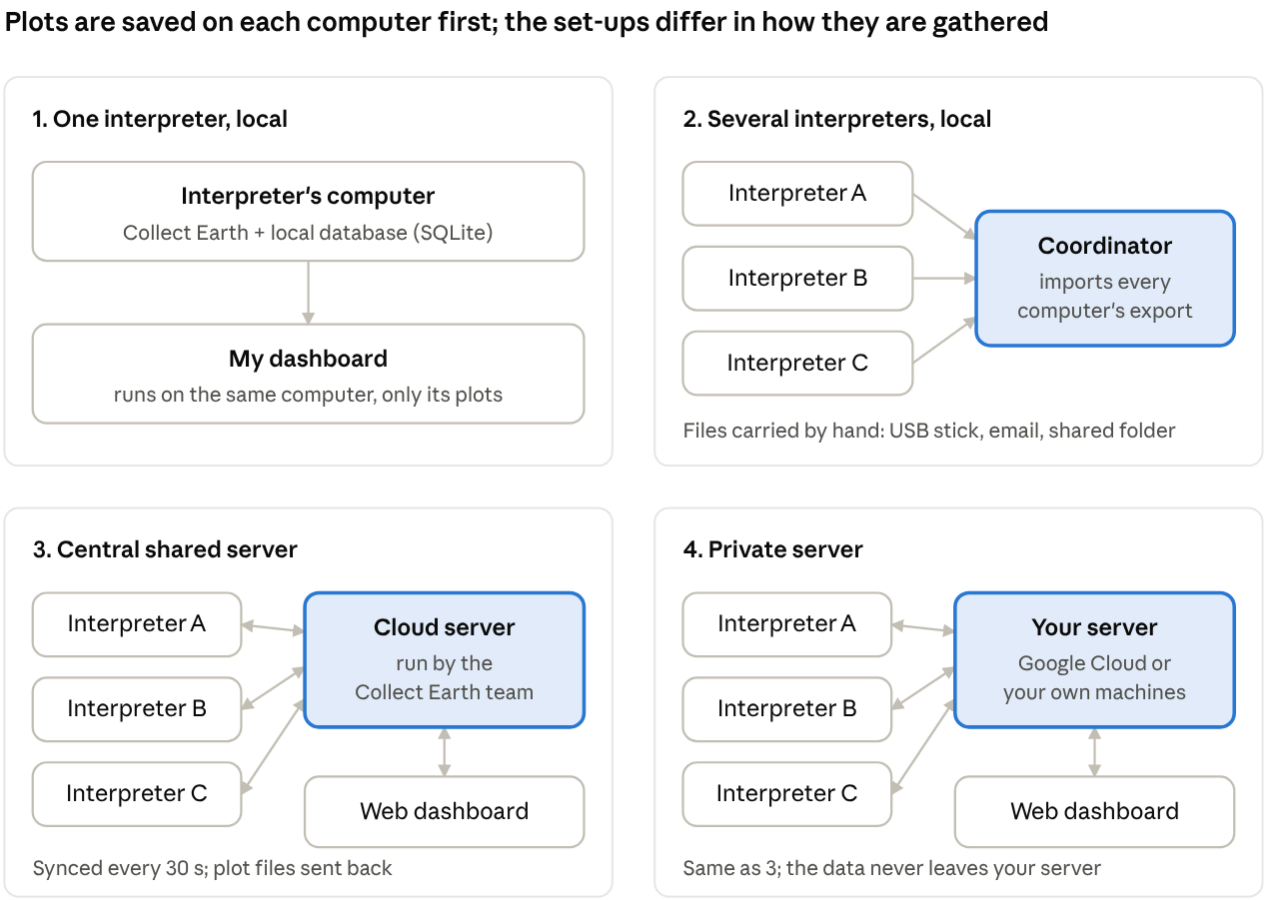
Collect Earth data collection: four ways to set it up

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Collect Earth always saves each plot on the interpreter's own computer first. The four set-ups differ in how those plots are brought together, and how much a coordinator can see while the work goes on.

The four set-ups at a glance



how the plots travel · 4 set-ups

In set-ups 1 and 2 nothing leaves the computers unless someone carries an export; in 3 and 4 every save travels to the server by itself, and plot files come back the same way.

	1. One interpreter, local	2. Several interpreters, local	3. Several interpreters, central shared server	4. Several interpreters, private server
Typical use	A study, a pilot, one expert	A small team in one office, no internet	Most team projects	Organisations that must keep data in-house
Where the plots are saved	The computer's own database (SQLite)	Each computer's own database, or one shared PostgreSQL database	Each computer's database, synced to the cloud server	Each computer's database, synced to your server
Internet needed	Only for satellite imagery	Only for imagery (a local network for shared PostgreSQL)	Imagery, plus now and then to sync	Imagery, plus a connection to your server
How plots reach the coordinator	They are already there	Export on each computer, import on the coordinator's	Automatically, every 30 seconds	Automatically, every 30 seconds
Giving out plots	Not needed	By hand: one plot file per person	Dashboard: whole files or divided automatically	Dashboard, as in set-up 3
Live progress dashboard	My dashboard (yours only)	My dashboard on each computer (its own plots only)	Full web dashboard for the team	Full web dashboard for the team
Quality checks, training, time spent	Not available	Done by hand	Built in	Built in
Who runs the server	Nobody	Nobody (or IT, for PostgreSQL)	The Collect Earth team	You
Cost	Free	Free	Free	About 10 to 15 US$ a month on Google Cloud

How to choose

- **Working alone?** Use set-up 1. You can still move to a cloud project later.
- **A team, and the internet is reliable at least once a day?** Use set-up 3. It needs no installation beyond Collect Earth, and it gives the coordinator progress, quality checks, training and the time spent per plot.
- **A team, with a rule that data stays on your organisation's servers?** Use set-up 4. It works exactly like set-up 3, on a server you install and pay for.
- **A team with no internet at all?** Use set-up 2, and plan regular exports from each computer. Expect more manual work and fewer checks.

Set-ups 3 and 4 still work offline: each interpreter keeps working, and the plots are sent when the connection comes back.

What every set-up has in common

In all four set-ups the interpreter uses the same Collect Earth desktop program and the same project file. What changes is only where the saved plots go next.

The project file (`.cep`)

A `.cep` file is a ZIP made with Collect Earth's survey designer. It holds the survey (the questions), the form, the plot file, the Google Earth template and the project settings. Open it with **File → Import CEP file**, or drop it on the Collect Earth window. Collect Earth unpacks it into its user folder and restarts with the new project.

The plot file (`.csv` or `.ced`)

One row per plot: the plot id, latitude and longitude, then any columns the survey expects. Change it in **Settings → Project → Path to ced/csv file with plot data**. The settings page shows how many plots it holds, its first rows and where the plots are.

Where the plots are saved

Setting (**Settings → Data storage**)	Where the data lives	For
SQLite (the default)	One file, `collectEarthDatabase.db`, in the user folder, shared by every project	Set-ups 1 and 2
PostgreSQL	A database server several computers can use at once	Set-up 2, on a local network

Setting (**Settings → Data storage**)	Where the data lives	For
Cloud project	The same SQLite file, synced to a server in the background	Set-ups 3 and 4 (set by **Join cloud project…**)

The user folder is `%APPDATA%\CollectEarth` on Windows, `~/Library/Application Support/CollectEarth` on macOS and `~/CollectEarth` on Linux. **Tools → Open data folder** opens it.

Your name

Collect Earth saves the interpreter's name (the *operator*) with every plot, so plots from different people can be told apart. It asks for it the first time (6 to 49 characters). Click the name in the top bar to change it; the change applies to later saves only. In a cloud project the name is the account name and cannot be changed.

Looking at the plots and filling in the form

Settings → Imagery and viewer → Plot viewer offers three ways:

- **Google Earth Pro (KML)**: the classic way; the form opens in Google Earth's balloon, or in the browser if chosen.
- **Plots map window**: a map of the plots in Collect Earth's own window, with the form beside it.
- **Plots map + Google Earth Web**: the plots map with Google Earth on the web for imagery.

The form saves every change by itself ("All changes saved"). A plot counts as done only when the interpreter presses **Submit**. Changing a submitted plot un-submits it until it is submitted again.

Plot colour	Status
Red	Not started
Yellow	In progress (saved, not submitted)
Green	Submitted

After **Submit**, Collect Earth moves to the next unfinished plot. **Next unfinished plot** and the **To do / Done** filter help find the rest.

Time spent on each plot

Collect Earth counts the time each plot's form is open and in use, leaving out pauses over 5 minutes. It is kept in the local database in every set-up, and shown to coordinators in set-ups 3 and 4.

Set-up 1: one interpreter, local storage

One person installs Collect Earth, opens the project file and assesses the plots. Everything stays in the SQLite file on that computer. Only the satellite imagery needs the internet.

Getting started

1. **Install Collect Earth** for your system (Windows, macOS or Linux). It brings its own Java, and the Node program that runs *My dashboard*.
2. **Type your name** when asked. It is saved with every plot.
3. **Open the project: File → Import CEP file**, choose the `.cep`, then **Import the project**. Collect Earth restarts with it.
4. **Check the settings:** the plot file (**Settings → Project**), the plot viewer (**Imagery and viewer**), and that **Data storage** is *SQLite*.

Assessing plots

1. Pick a plot on the map (or in Google Earth Pro). Its form opens.
2. Look at the imagery and answer the questions. Each answer is saved at once.
3. Press **Submit** when the plot is complete. The plot turns green and the next unfinished plot opens.
4. Repeat. The sidebar shows *{saved} of {total} completed*.

Following your progress: My dashboard

Tools → My dashboard opens a dashboard in your browser that runs on your own computer. It starts by itself when the project is loaded, and gets every new save within seconds.

- **Tabs:** Progress, Map, Data, Quality, Activity and Settings (a deadline and a target).
- **It sees only this computer's database.** Nothing is sent anywhere.
- **Not in a local dashboard:** invitations, giving out plots, automatic checks and reference plots, which need other people.

Keeping the data safe

- **Automatic backup:** with **Back up the database automatically** on (the default), Collect Earth zips the SQLite file into `backupSqlite` in the user folder each time it closes, and keeps the newest 10.
- **Your own copy: Data Import/Export → Download data to CSV (Excel)** opens the **Export data** dialog. Export **XML (compressed)** now and then, and keep it somewhere else (a USB stick, a shared drive).

Getting the results out

Export format	Use it for
CSV (Excel)	Analysis in a spreadsheet or R; tick *Include labels for code attributes* for readable answers
XML (compressed)	Moving plots to another Collect Earth (it is the only format another Collect Earth can import)
Collect backup	Opening the data in Open Foris Collect
KML	Seeing the plots and answers in Google Earth

Tools also has **Start SAIKU Analysis** for tables and charts, and **Generate LULUCF package** for greenhouse gas reporting.

Checking nothing was missed

Utilities → Find missing plots in database compares a plot file with the database and lists plots that are not submitted, with **Save the missing plots as CSV**.

Set-up 2: several interpreters, local storage

Each interpreter works as in set-up 1, and a coordinator brings the plots together by hand. There are two ways to do it: each computer keeps its own database and sends exports, or every computer writes into one shared PostgreSQL database on the office network.

Way A: each computer its own database (works with no network)

Coordinator, before the work starts

1. **Give everyone the same `.cep` file.** Every computer must use the same survey, or the exports cannot be merged.

2. **Split the plots. Utilities → Divide large CSV plot files** cuts one plot file into a number of files (up to 500), optionally in random order or by the values of a column (for example a province). Files go to `div_<N>` in the folder you choose.
3. **Hand out one file per person**, and keep a list of who has which file. Each interpreter sets it in **Settings → Project**.
4. **For quality checks**, give some plots to two people on purpose, and note which. Collect Earth does not pick them for you in this set-up.

Each interpreter

1. Uses their own name (Collect Earth asks for it) so their plots can be told apart.
2. Assesses their plots as in set-up 1.
3. Exports regularly (once a day or once a week): **Export data → XML (compressed, to share with other Collect Earth installations)**, choosing **Records modified since the last export** to send only what is new.
4. Sends the `.zip` to the coordinator (USB stick, email, shared folder).

Coordinator, bringing the plots together

1. **Back up first:** export your own database as XML before importing.
2. **Data Import/Export → Import new data from ZIP (compressed XML)**, choose one or more `.zip` files, then **Read the files**.
3. Choose **Import only fully assessed plots (green tick)** (the default) or **Import ALL plots in the file**.
4. **When a plot is already in your database**, the dialog lists both versions with their date and how complete each is. For each file choose **Keep the records of my database** (the default) or **Overwrite them with the records of the file**.
5. Check the summary: *records imported, overwritten, kept*.
6. Run **Utilities → Find missing plots in database** with the original plot file to see what is still missing.

Watch out

- **The same plot from two people:** records are identified by the plot, not by the person, so importing two interpreters' versions of one plot keeps only one. For a quality check, keep the second person's export apart and compare it in a spreadsheet.
- **Nothing shows progress across the team** until the exports arrive. *My dashboard* on each computer shows that computer's plots only.
- **No tool renames an operator.** Agree on names before starting.

Way B: one shared PostgreSQL database (office network)

IT installs a PostgreSQL server that every computer can reach. Then on every computer:

1. **Settings → Data storage → PostgreSQL**, with **DB host**, **DB port** (5432), **DB Name**, **User Name** and **Password**. Collect Earth restarts.
2. Open the same `.cep` and give each person their own plot file, as in way A.

Everyone's saves go straight into one database, and plot colours on each map update every few seconds as others work. There are no exports to carry.

Watch out

- **If the database cannot be reached at start**, Collect Earth silently falls back to the computer's own SQLite file and says so in a notice. Plots saved then are **not** in the shared database: export and import them as in way A.
- **Two people on the same plot overwrite each other**: the last save wins. Never give the same plot to two people in this way.
- **No automatic backup** is made of a PostgreSQL database: IT must back it up.
- **My dashboard** sends everything under the one name of the computer it runs on, so it cannot show who did what.

Set-up 3: several interpreters with a central shared server

Each interpreter works in Collect Earth as usual, and every plot is sent to Collect Earth Cloud, the free server run by the Collect Earth team. The coordinator follows and checks the work in a web dashboard. Nobody installs a server.

Who does what

Role on the dashboard	What they do	What they see
Coordinator	Creates the project, uploads the files, invites people, gives out plots, decides on differing answers, downloads the data	Everything, including names, quality per person, training scores and time spent per plot
Reviewer	Helps check quality	Progress, the map, plot pages, plots worth a look, calibration figures without names. Not the People tab or the time spent

Role on the dashboard	What they do	What they see
Collector (the interpreter)	Assesses plots in Collect Earth	**My work** only: their own plots, their own quality figures and their training

A project can have several coordinators. The last coordinator cannot be removed or demoted.

Coordinator: setting up the project

1. **Create an account and a project.** Open the server's address, sign up, then **My projects → New project**. The project page shows a checklist of the next steps.
2. **Upload the project file (`.cep`).** Use the **Project file** tab. This is the same file you would open in Collect Earth: the survey, its form and its settings. If no plots have been collected yet, the server adds a hidden `round` key to a survey identified by plot id only, so blind quality checks can be told apart from first assessments later.
3. **Upload the plot files (CSV).** Use the **Plot files** tab. Each file is the same CSV Collect Earth reads: a plot id, then latitude and longitude, then any extra columns.
4. **Give out the plots.** For each plot file, choose one of:
   - **Whole file** to one or more people.
   - **Divided evenly** between chosen people, in file order or mixed, with an optional overlap percentage. The overlap gives some plots to a second person as a blind quality-check copy.
   - **Divided by a column**, for example by province or stratum, with people chosen for each value. Later, **Move unfinished plots** hands one person's remaining plots to someone else without touching what they already did.
5. **Invite people.** On the **People** tab, make an invitation link with a role (collector, reviewer or coordinator), how many people may use it and when it expires. Send the link by email or chat. One link with several uses works for a whole training room.
6. **Optional settings.**
   - Progress: a deadline, a target number of plots, and after how many quiet days a person is flagged.
   - Automatic checks: a share of each person's first plots (or a fixed first number) is copied to chosen checkers as blind second assessments.
   - Reference plots: plots an expert assesses, which every collector also gets blind, to measure accuracy.

- Training: calibration and learning plot sets, kept apart from the real data (see below).

Interpreter: joining and working

1. **Install Collect Earth** (the release after 1.23.19 or later is needed for blind checks).
2. **Join.** Click the invitation link. The dashboard shows the steps: in Collect Earth choose **Join cloud project…**, paste the link, check the project name and role it shows, then create an account (or log in with an existing one).
3. Collect Earth **downloads the project file, switches to cloud mode and restarts**. The window title ends with *[Cloud project]*.
4. **Choose your plot file.** In **Settings**, the **Assigned plot file** list shows the files the coordinator gave you. Pick one. **Refresh assignments** fetches new ones.
5. **Assess plots** as in set-up 1. The operator name is your account name and cannot be changed in a cloud project.
6. **Sync happens by itself.** Every save is queued and sent in the background every 30 seconds. A full check against the local database runs every 10 minutes, so nothing is missed. Offline, the queue waits and retries, up to once an hour. **Cloud sync status…** shows plots waiting, failed and synced, and has a **Sync now** button.
7. **See your own progress** on the dashboard's **My work** page: your plots, your quality figures, your training sets and their feedback.

Coordinator: following the work

Dashboard tab	What it answers
Progress	How many plots are done, per day, with a forecast against the deadline, and who needs attention
People	Each person's plots, speed, last sync, agreement, and the **time spent on each plot** (quick plots flagged)
Map	Every plot coloured by status: not started, started, submitted, accepted, needs another look, answers differ
Quality	Agreement between first assessments and checks, per question and per person; plots worth a look (very fast, same answers as before, required answers missing, edited after a decision); the review queue; decisions; guidance notes; reference plots; optional AI assistant

Dashboard tab	What it answers
Training	Calibration and learning plot sets, golden answers, scores, printable reports
Activity	A log of what happened: plots submitted, files changed, people joined
Data	Downloads: a CSV of all answers (with the coordinator's final decision per plot), and a Collect backup to open in Collect or Collect Earth

How the records are kept apart

Each record is stored by plot and by person. Two interpreters on the same plot never overwrite each other. A newer save of the same person's plot replaces the older one, and a deletion can be undone by saving again. Training plots go to their own table and never appear in progress, quality figures or downloads.

Time spent on each plot

Collect Earth counts how long each plot's form was open and in use, from opening to submitting, summed over every opening. Pauses longer than 5 minutes are left out. Coordinators see each person's usual time, and plots done in under a quarter of the team's median are listed as **quick plots** (once 20 plots have a time). Reviewers and collectors do not see these times.

Set-up 4: several interpreters with a private server

A private server is your own copy of Collect Earth Cloud. Once it is running, coordinators and interpreters work exactly as in set-up 3; only the server's address is different. Nobody else, including the Collect Earth team, can see the data.

Two ways to install it

	Google Cloud, guided (`SELF-HOSTING.md`)	Any Docker host (`DEPLOY.md` , option B)
Skills needed	None: a button, a panel, one command	A system administrator

	Google Cloud, guided (`SELF-HOSTING.md`)	Any Docker host (`DEPLOY.md` , option B)
What you need first	A Google account and a Google Cloud billing account	A server that runs Docker, a PostgreSQL database with backups, and HTTPS in front
Time	About 20 minutes, most of it waiting for the database	Depends on your infrastructure
Cost	About 10 to 15 US$ a month (database 8 to 10, disk and backups 2 to 3, the server almost free)	Your own hosting costs
Backups	Every night, the last 7 kept, plus recovery to any minute of the last 7 days	Yours to arrange

Installing on Google Cloud, step by step

1. Open the **Open in Cloud Shell** button in `SELF-HOSTING.md` . Tick **Trust repo** and **Confirm**.
2. Follow the panel on the right: pick or create a Google Cloud project, and check billing is on.
3. Run `bash deploy/gcp/deploy.sh` . Choose the region nearest your team (Enter for Belgium, `europe-west1`) and confirm.
4. Wait for the 8 steps. The database takes 10 to 15 minutes, building the server 3 to 6 more.
5. **Write down the dashboard address and the setup code** it prints. The code is needed once; `deploy.sh --show-setup-code` shows it again.

Installing on your own servers

Build the image (`docker build -t collect-earth-cloud .` in `collect-earth-cloud`) and run it with at least:

Setting	What it is
`STORE_BACKEND=postgres`	Keep data in PostgreSQL (the default memory store loses everything on restart)
`MIGRATION_DATABASE_URL`	The database owner, used to create and update the tables

Setting	What it is
`DATABASE_URL`	A separate, less powerful role the server runs as, so row-level security keeps each project's data apart
`SETUP_CODE`	The one-time code for the first administrator
`PUBLIC_BASE_URL`	Your HTTPS address, used in invitation links
`MIGRATE_ON_BOOT=true`	Apply database updates when the server starts

Optional: `SMTP_URL`, `MAIL_FROM` and `LINK_SECRET` for emails (weekly summaries, address confirmation, "Forgot your password?"); `TELEMETRY=1` to send error reports and usage counts (off by default on a private server); the AI assistant settings. The server speaks plain HTTP, so put HTTPS in front of it (a load balancer, nginx or Caddy).

First steps on a new server

1. **Setup.** Open the address, type the setup code, and choose the server administrator's user name and password.
2. **Server settings** (the **Server** page):
   - *Anyone can create an account*: leave **off** for a private server. People then only get accounts through project invitations.
   - *Who can create projects*: only server administrators, or everyone with an account.
   - Optional limits per account and per project, a server name, terms of use and a contact email.
3. **Create the first project** and continue as the coordinator in set-up 3. The server administrator sees no project's data unless they are also a member of it.
4. **Forgotten passwords:** without email, the administrator makes a reset link on **Server → Accounts** and sends it. With email set up, people use **Forgot your password?** themselves.

Looking after it

- **Updating:** run the same `deploy.sh` command again. It rebuilds and switches in about 5 minutes; the data stays and users do not notice.
- **Your own address:** `deploy.sh --domain cloud.example.org`, then add the DNS records it prints.
- **Restoring a backup** replaces all current data, so download each project's data first.
- **Deleting everything** (to stop paying): delete the Google Cloud project. Google keeps it 30 days in case you change your mind.

- **If something fails:** `gcloud run services logs read collect-earth-cloud --region <your region> --limit 50` shows the server's log.

The interpreters' side

Nothing changes for them: they click the invitation link from your server, choose **Join cloud project…** in Collect Earth, and sync with your server from then on. The link carries the server's address, so they never type it.

Day to day, problems and questions

An interpreter's day

	Set-ups 1 and 2 (local)	Set-ups 3 and 4 (server)
Start	Open Collect Earth; the project opens where you left it	The same; log in if Collect Earth asks
Plots to do	Your plot file, in **Settings → Project**	**Settings → Assigned plot file**; **Refresh assignments** for new ones
While working	Answer, then **Submit** each plot	The same; plots are sent by themselves
End of day	Set-up 2, way A: export *Records modified since the last export* and send it	Nothing; check **Cloud sync status…** shows nothing waiting if you will be offline for long
See your progress	**Tools → My dashboard**	**My work** on the dashboard
Quit	**Quit** at the right of the top bar (the automatic backup runs when Collect Earth closes). Closing the window asks whether to quit or open it again	The same

Common problems

What you see	Why	What to do
A notice that PostgreSQL could not be reached	The database server or network is down	Collect Earth is using its own SQLite file meanwhile: export those plots later, or fix the connection and restart
Cloud sync status shows plots waiting or failed	No internet, or the server is down	Nothing: they are retried by themselves. **Sync now** tries at once
"You are not logged in to the cloud project"	The password changed, or the session ended	Log in again; nothing saved locally is lost
"The cloud server assigns you no plot file in this project yet"	The coordinator has not given you plots	Ask the coordinator, then **Refresh assignments**
An imported plot did not appear	It was not submitted, or your database already had it and *Keep* was chosen	Import again with **Import ALL plots** or **Overwrite**
A plot is yellow, not green	It was changed after it was submitted	Open it and press **Submit** again
Collect Earth seems to stop after changing the project or a setting	It restarts itself, which takes a little while	Wait: a small Collect Earth window shows each step until the new window opens

Questions

- **Can I move from local work to a server later?** Yes. The coordinator uploads the same `.cep` to the server, and plots already assessed can be brought in by importing each computer's XML export into one Collect Earth that has joined the project.
- **Does a server project still work without internet?** Yes. Plots are saved locally and sent when the connection returns.
- **Can two people assess the same plot?** On a server, yes: each person's version is kept and compared in the quality figures. Locally, only one version of a plot can be kept per database.
- **Who can see the time each plot took?** Only coordinators on a server (set-ups 3 and 4).

- **Where is the data on my computer?** In the user folder: **Tools → Open data folder**.